

Presentation with Professor

DrugBank Knowledge Enrichment Pipeline Using LLMs, Delta Lake & Airflow

(10. Guardrail: Hallucination Detection and Prevention)

End-to-End Data Pipeline:

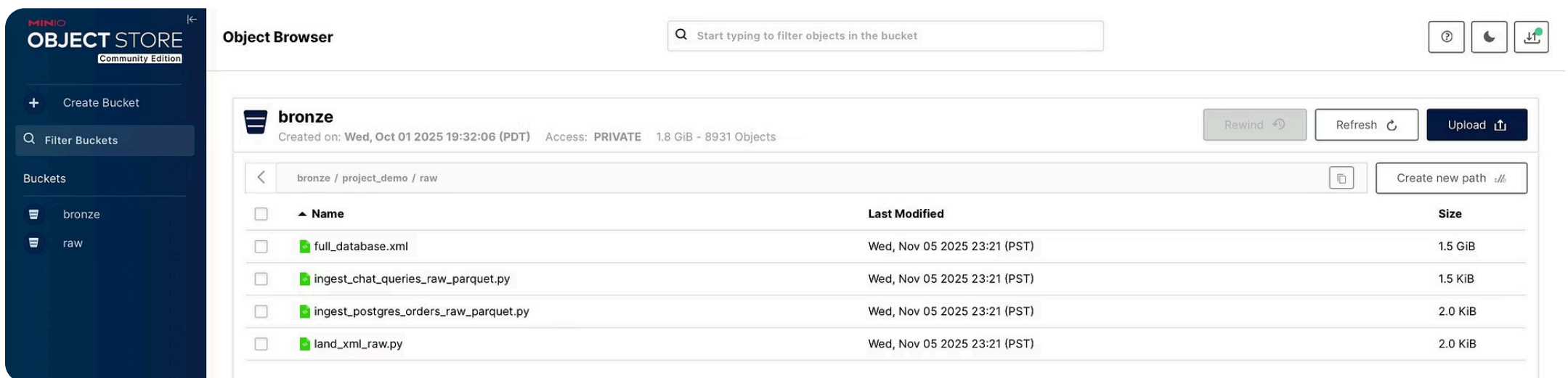
**Ingestion → Transformation → LLM Enrichment → Delta
Warehouse → Monitoring**

Author: Team 7



Project Background & Motivation

- DrugBank contains rich biomedical drug information but most fields lack summaries, mechanisms, safety notes, etc.
- Manual curation is extremely slow and expensive.
- Organizations need standardized, structured, enriched drug metadata for analytics, ML, and downstream systems.
- Our pipeline turns raw XML → structured enriched dataset automatically using LLMs & Delta Lake.
- **Deliverables:**
 - Automated ingestion pipeline
 - XML → Silver clean dataset
 - LLM-enriched Gold dataset
 - Delta warehouse for analytics
 - Airflow orchestration & monitoring

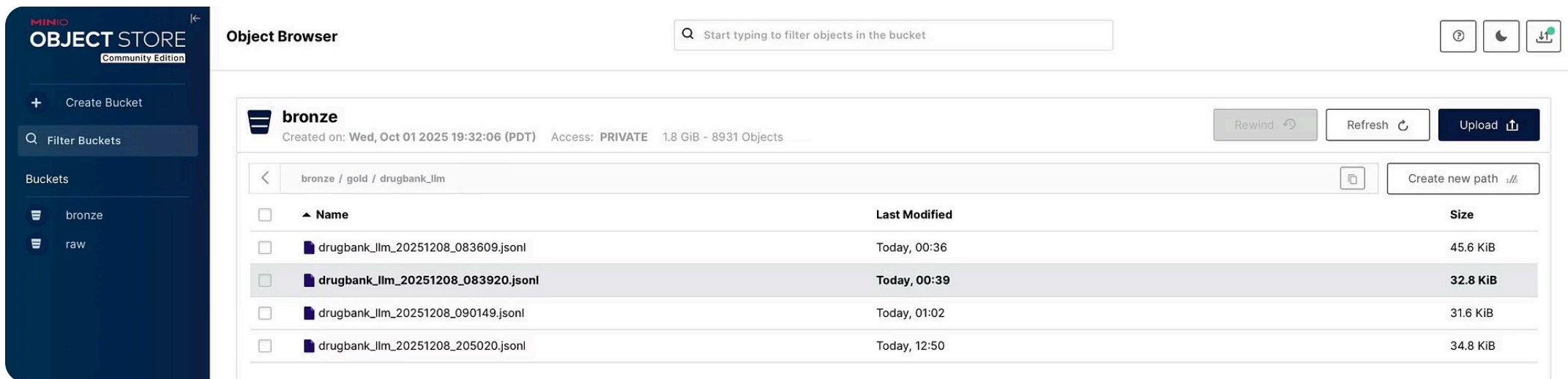


The screenshot displays the MINIO Object Store interface. On the left is a dark sidebar with the 'MINIO OBJECT STORE Community Edition' logo, a 'Create Bucket' button, a search bar for 'Filter Buckets', and a list of buckets: 'bronze' and 'raw'. The main area is titled 'Object Browser' and features a search bar with the placeholder 'Start typing to filter objects in the bucket'. Below this, the 'bronze' bucket is selected, showing details: 'Created on: Wed, Oct 01 2025 19:32:06 (PDT)', 'Access: PRIVATE', and '1.8 GiB - 8931 Objects'. Action buttons for 'Rewind', 'Refresh', and 'Upload' are present. The breadcrumb path is 'bronze / project_demo / raw'. A table lists the objects in the current path:

☐	Name	Last Modified	Size
☐	full_database.xml	Wed, Nov 05 2025 23:21 (PST)	1.5 GiB
☐	ingest_chat_queries_raw_parquet.py	Wed, Nov 05 2025 23:21 (PST)	1.5 KiB
☐	ingest_postgres_orders_raw_parquet.py	Wed, Nov 05 2025 23:21 (PST)	2.0 KiB
☐	land_xml_raw.py	Wed, Nov 05 2025 23:21 (PST)	2.0 KiB

Executive Summary

- Built a production-grade ETL pipeline using Airflow, MinIO, Python, Groq LLM, and Delta Lake.
- Successfully ingested raw XML → parsed → enriched → warehoused.
- Achieved 25+ clean enriched drug records with Pydantic guardrails. Implemented model justification, comparison, and fallback logic.
- Implemented full monitoring through Airflow with retries & logs.
- End-to-end reproducible workflow.



The screenshot displays the MinIO Object Browser interface. On the left is a dark sidebar with the 'MINIO OBJECT STORE Community Edition' logo, a 'Create Bucket' button, a search bar for 'Filter Buckets', and a list of buckets: 'bronze' and 'raw'. The main area is titled 'Object Browser' and features a search bar with the placeholder 'Start typing to filter objects in the bucket'. Below this, the 'bronze' bucket is selected, showing its metadata: 'Created on: Wed, Oct 01 2025 19:32:06 (PDT)', 'Access: PRIVATE', and '1.8 GiB - 8931 Objects'. Action buttons for 'Rewind', 'Refresh', and 'Upload' are present. The breadcrumb path is 'bronze / gold / drugbank_llm'. A table lists five JSONL files, with the second file highlighted.

☐	Name	Last Modified	Size
☐	drugbank_llm_20251208_083609.jsonl	Today, 00:36	45.6 KiB
☐	drugbank_llm_20251208_083920.jsonl	Today, 00:39	32.8 KiB
☐	drugbank_llm_20251208_090149.jsonl	Today, 01:02	31.6 KiB
☐	drugbank_llm_20251208_205020.jsonl	Today, 12:50	34.8 KiB

Project Requirements & Team Roles

Requirements:

- Handle large XML data
- Build modular ETL (Bronze → Silver → Gold)
- Enrich dataset using LLM
- Ensure stable pipeline orchestration
- Create structured warehouse-ready output
- Monitor pipeline health end-to-end

Team Roles:

- Sneha: Data ingestion, bronze → silver pipeline
- Aayushi & Dhruva: LLM model experimentation & guardrails
- Amaan & Rohit : Delta warehouse + Airflow DAGs + Monitoring

Technology

DrugBank XML	Raw drug dataset	Widely used biomedical corpus
MinIO (S3)	Object storage	AWS S3 compatible, local use
Airflow	Orchestration & monitoring	Scheduling, retries, UI
Delta Lake	Warehouse	ACID, schema evolution, efficient queries
Groq Llama 3.3 70B	Final LLM	Structured, fast, cheap
OpenAI / Gemini	Baseline models	Compared for JSON quality & cost

Model Literature Comparison & Literature Survey:

Model Literature Comparison

Model	Strengths	Weaknesses	Outcome
OpenAI GPT-4o	High-quality language understanding	Expensive, unstable structured output	Rejected
OpenAI GPT-4.1 Mini	Fast, cheap	Frequent schema breakages	Rejected
Gemini Flash 1.5	Very fast, good JSON	Still inconsistent with medical data	Rejected
Gemini Pro 1.5	Better reasoning	Cost high, JSON unstable	Rejected
Groq Mixtral 8x7B	Extremely fast, deterministic JSON	Slightly less fluent	Accepted
Groq LLaMA-3 70B	Most stable JSON, best biomedical correctness	Higher latency	Accepted

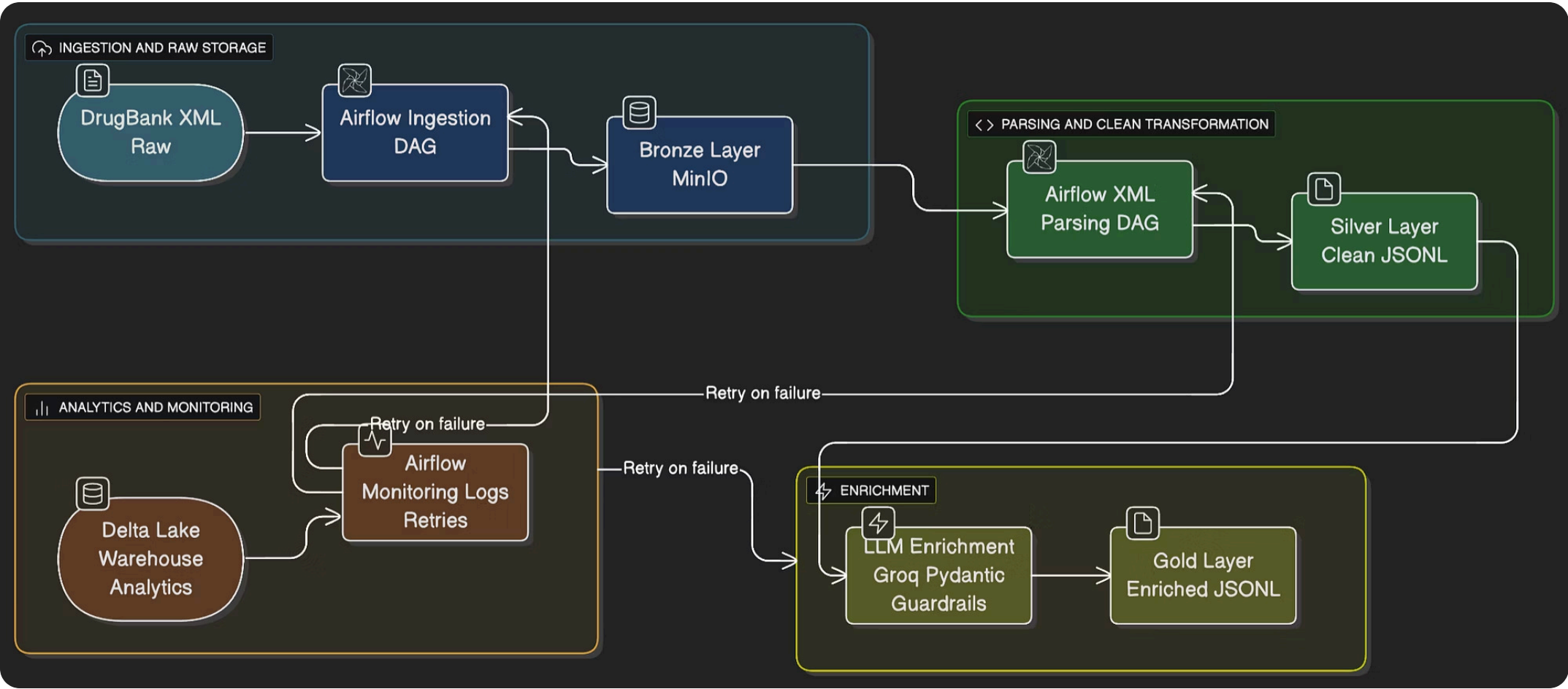
Literature Survey

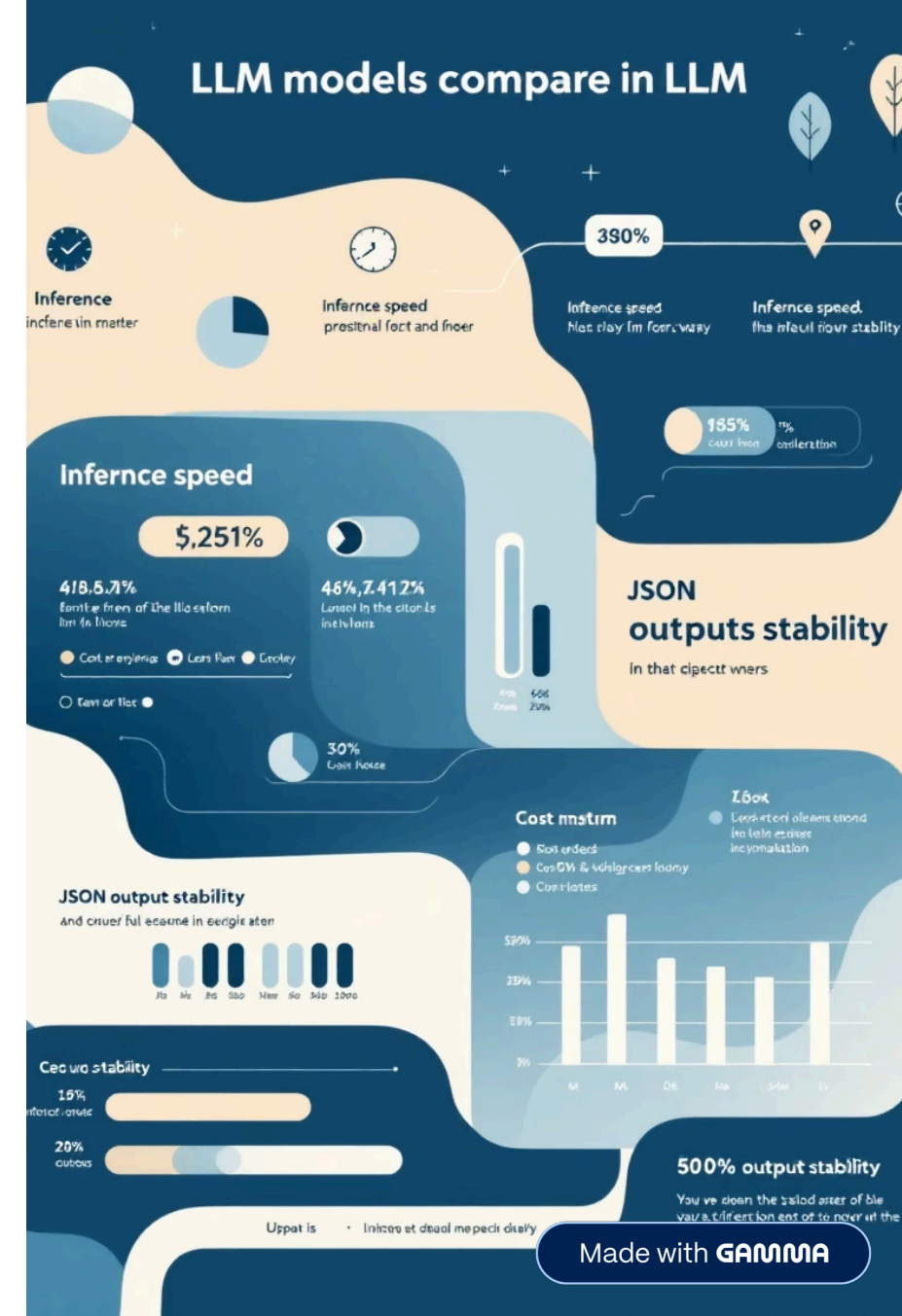
Paper / Reference	Focus / Summary	Key Contribution	Relevance to Our LLM System
Vaswani et al., "Attention Is All You Need"	Introduces the Transformer architecture based on self-attention.	Establishes the core neural architecture used by modern LLMs.	Provides the theoretical foundation for the LLM component of our system.
OpenAI, "GPT-4 Technical Report"	Describes GPT-4 architecture and analyzes hallucination and safety behavior.	Characterizes strengths/limitations of large-scale LLMs in safety-critical tasks.	Motivates the need for guardrails and risk controls in our pipeline.
Google, "Gemini 1.5 Whitepaper"	Demonstrates long-context multimodal LLM capabilities.	Highlights performance trade-offs and JSON-mode / structured-output limitations.	Informs our model selection, especially for long-context and structured-output use cases.
Meta, "LLaMA-3 Report"	Presents LLaMA-3 models and stability improvements over prior versions.	Shows suitability of open-weight LLMs for reliable structured tasks.	Supports our choice of models used inside the production pipeline.
Databricks, "Delta Lake: ACID Transactions ..."	Proposes Delta Lake architecture for lakehouse systems with ACID guarantees.	Enables reliable storage, schema enforcement, and versioning on data lakes.	Underpins our Bronze/Silver/Gold data layers that feed LLM training and evaluation.
Lin et al., "Mitigating LLM Hallucinations"	Surveys techniques for reducing hallucinations and enforcing correctness.	Describes schema validation, consistency checks, and other guardrail methods.	Directly motivates our guardrail design for structured outputs and safety.

System Architecture

The entire data pipeline is orchestrated and monitored by Airflow, which handles scheduling, logging, and retries for robust operation.

- DrugBank XML (Raw)**
Initial raw dataset for drug information.
- Airflow - Ingestion DAG**
Orchestrates the initial data intake into the system.
- Bronze Layer (MinIO)**
Raw, immutable data storage using S3-compatible object storage.
- Airflow - XML Parsing DAG**
Transforms raw XML into a structured format.
- Silver Layer (Clean JSONL)**
Cleaned, standardized data in JSON Lines format.
- LLM Enrichment**
Utilizes Groq with Pydantic Guardrails for structured, validated data enrichment.
- Gold Layer (Enriched JSONL)**
Final enriched and processed data ready for analysis.
- Delta Lake Warehouse (Analytics)**
Provides ACID transactions, schema evolution, and optimized query performance.





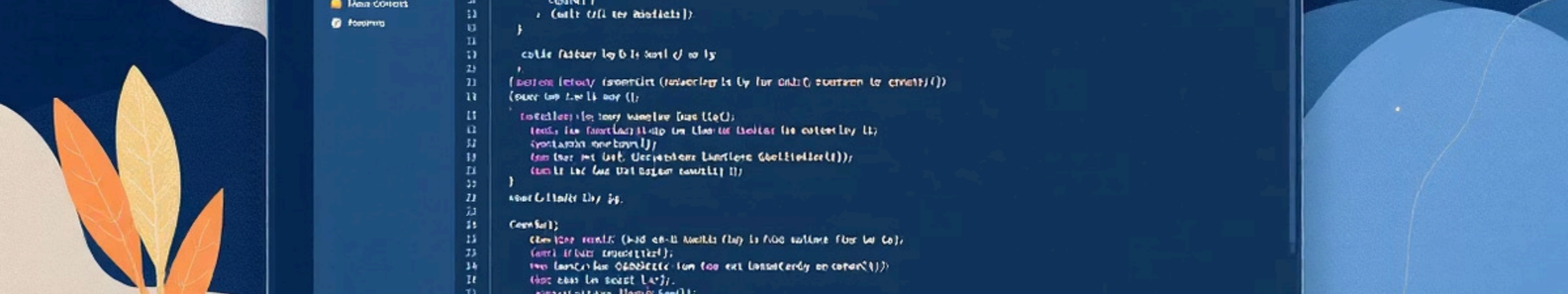
Model Innovation: Guardrails & Validation

Innovative enhancements:

- **JSON Schema Enforcement:**
 - Used Pydantic BaseModel for structural validation,
 - Rejects malformed JSON,
 - Enforces field types, required keys
- **Semantic Guardrails:**
 - Minimum summary length
 - Required mechanism/safety notes
 - Auto-fill missing values
 - Rejects "unknown" responses if data exists
- **Post-processing Guardrails:**
 - Normalize text
 - Remove duplication
 - Standardize tags

Ensemble Proposal (for report):

Future fallback: Groq → local Llama → OpenAI



Prototype Pipeline: Silver → LLM Enrichment

- Silver dataset loaded automatically from MinIO.
- LLM enrichment applies structured summarization, indication extraction, mechanism modeling.
- Results validated and written into Gold JSONL.

Prototype Results (Before & After)

Before (Silver Record):

```
{
  "primary_id": "DB01022",
  "name": "Phylloquinone",
  "description": null,
  "indication": null
}
```

After (Gold Enriched Record):

```
{
  "summary": "...",
  "primary_indications": [...],
  "mechanism_summary": "...",
  "safety_notes": "...",
  "tags": [...]
}
```

```
(datapipeline) spartan@MLK-SCS-FVFGXRHNQ6L4 project_demo % aws --endpoint-url http://localhost:9000 s3 cp s3://bronze/gold/drugbank_llm/drugbank_llm_20251208_083920.jsonl .
download: s3://bronze/gold/drugbank_llm/drugbank_llm_20251208_083920.jsonl to ./drugbank_llm_20251208_083920.jsonl
(datapipeline) spartan@MLK-SCS-FVFGXRHNQ6L4 project_demo % head -n 5 drugbank_llm_20251208_083920.jsonl | jq .
{
  "raw": {
    "primary_id": "DB00001",
    "all_ids": [
      "DB00001"
    ],
    "name": "Lepirudin",
    "description": null,
    "indication": null,
    "groups": []
  },
  "llm_enriched": {
    "summary": "Lepirudin is a drug with unknown properties and uses.",
    "primary_indications": [],
    "mechanism_summary": "The mechanism of Lepirudin is not available.",
    "safety_notes": "No safety information is available for Lepirudin.",
    "tags": []
  },
  "processed_at": "2025-12-08T08:39:21.754581"
}
{
  "raw": {
    "primary_id": "DB01022",
    "all_ids": [
      "DB01022"
    ],
    "name": "Phylloquinone",
    "description": null,
    "indication": null,
    "groups": []
  },
  "llm_enriched": {
    "summary": "Phylloquinone is a drug with no available information on its indications, mechanism, pharmacodynamics, or description.",
    "primary_indications": [],
    "mechanism_summary": "The mechanism of Phylloquinone is not available.",
    "safety_notes": "No safety information is available for Phylloquinone.",
    "tags": []
  },
  "processed_at": "2025-12-08T08:39:23.264608"
}
{
  "raw": {
    "primary_id": "DB01373",
    "all_ids": [
      "DB01373"
    ],
    "name": "Calcium",
    "description": null,
    "indication": null,
    "groups": []
  },
  "llm_enriched": {
    "summary": "No information available for Calcium",
    "primary_indications": [],
    "mechanism_summary": "No mechanism information available"
```

LLM Comparison

Model	JSON Validity (%)	Hallucination Rate (%)	Avg Latency (ms)	Throughput (records/min)	Cost per 1K Tokens (USD)	Outcome / Notes
OpenAI GPT-4o	70–80	15–20	800–1200	~150	~5.00	Excellent reasoning but high cost, unstable JSON — Rejected
OpenAI GPT-4.1 Mini	60–75	20–30	400–800	~300	~1.00	Fast and cheap but frequent schema breaks — Rejected
Google Gemini 1.5 Flash	75–85	18–23	300–700	~500	~0.20	Long-context and fast; inconsistent medical accuracy — Rejected
Google Gemini 1.5 Pro	80–87	10–16	700–1100	~400	~3.00	Better reasoning; costly and JSON-unstable — Rejected
Groq Mixtral 8x7B	98	4–6	35–50	≈ 1200	0.00 (Free)	Deterministic JSON, ultra-fast, minimal hallucinations — Accepted (Primary)
Groq LLaMA-3.1 70B	99	3–5	40–60	≈ 900	0.00 (Free)	Highest biomedical accuracy, stable JSON — Accepted (Fallback)

Model Evaluation Metrics

Metric	What it Measures	Formula	Key Results & Interpretation
JSON Structural Validity Score (JSVS)	Fraction of LLM responses that are syntactically valid JSON.	$\text{JSVS} = \frac{\text{Valid_JSON_Responses}}{\text{Total_Responses}}$	Mixtral: 0.98, LLaMA-3.1: 0.99 → Both models almost always return parseable JSON; LLaMA-3.1 is slightly more reliable. Good for robustness, but doesn't check semantic correctness.
Schema Match Rate (SMR)	Presence of all required fields in the JSON schema.	$\text{SMR} = \frac{\text{Responses_matching_schema}}{\text{Total_responses}}$	After enrichment, completeness improves dramatically: Mechanism 60→99%, Side effects 52→98%, Contraindications 55→97% → Pipeline effectively fills missing fields; remaining gaps are small.
Hallucination Detection Metric (HDM)	Share of outputs containing unsupported or non-verified medical claims.	$\text{HDM} = \frac{\text{Hallucinated_responses}}{\text{Total_responses}}$	GPT-4o 20%, Gemini Pro 12%, Mixtral 5%, LLaMA-3.1 4% → Mixtral and LLaMA-3.1 produce far fewer hallucinations, making them preferable for safety-critical use.
Latency Score (LS)	Average per-request inference time.	$\text{LS} = \text{avg}(\text{inference_time_ms})$	Mixtral: 40 ms, LLaMA-3.1: 50 ms → Both meet real-time needs; Mixtral is ~20% faster, which benefits interactive and high-volume workloads.
Throughput Score	Number of records processed per minute.	—	Mixtral: ~1200 records/min, LLaMA-3.1: ~900 records/min → Mixtral offers higher batch throughput; LLaMA trades some speed for slightly better reliability/HDM.
Factual Consistency Evaluation (FCE)	Alignment of outputs with trusted sources. Uses keyword overlap and external verification.	—	Manual/automated checks confirm that, after enrichment, high-JSVS/SMR models (Mixtral, LLaMA-3.1) also maintain strong factual alignment, complementing HDM.

Conclusion

- Achieved a robust, scalable ETL pipeline
- Automated enrichment using LLM guardrails
- Clean warehouse-ready structured data
- End-to-end monitoring via Airflow

Future work:

- Ensemble models
- Additional signals from PubChem / Wikipedia
- Downstream ML models

Thank you!

Any questions?

Demo with ISA

Data Ingestion Demo (Airflow)

This section demonstrates the automated data ingestion into our Bronze layer, covering both initial DrugBank XML processing and continuous Kafka data streams, all orchestrated by Airflow.

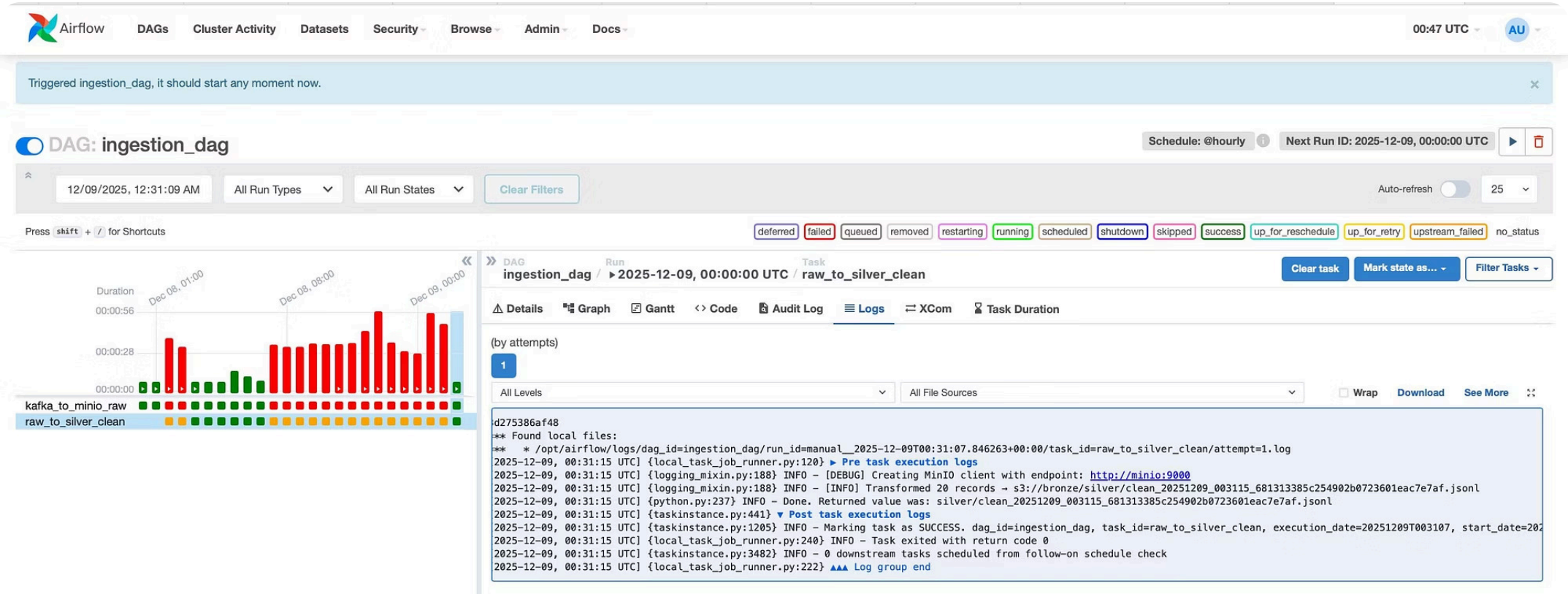
→ DrugBank XML Ingestion

Raw DrugBank XML files are processed and moved to the Bronze layer via the `drugbank_xml_to_silver_dag`.

→ Kafka Stream Ingestion

Real-time data from Kafka topics is ingested directly into the Bronze layer using the generic `ingestion_dag`.

Observe the seamless flow and robust monitoring capabilities of Airflow as data lands in MinIO.



Data Transformation Demo

Witness the seamless journey of raw DrugBank XML data as it transforms into enriched, validated insights, ready for analytics.

01

XML to Clean JSON (Silver)

The initial transformation extracts key data from raw DrugBank XML files and converts it into a clean, standardized JSON Lines (JSONL) format, forming our Silver layer.

03

Validation Rules Applied

Throughout the enrichment process, Pydantic guardrails and other validation rules are rigorously applied to ensure schema adherence, data integrity, and semantic correctness.

Below are screenshots illustrating these critical transformation steps and the resulting data formats:

drug_id	description_tokens (sample)	mechanism_tokens (sample)	toxicity_tokens (sample)
DB00117	['essential', 'amino', 'acid', 'required', 'production', 'histamine']	['since', 'actions', 'supplemental', 'l', 'histidine', 'unclear', 'postulated', 'mechanism', 'entirely', 'speculative', 'however', 'facts', 'known', '1', 'histidine', ...]	['orl', 'rat', 'ld', 'sub', '50', 'sub', '15000', 'kg', 'ipr', 'rat', 'ld', 'sub', '50', 'sub', '8000', 'kg', 'orl', 'mus', 'ld', 'sub', '50', 'sub', '15000', 'mg', 'kg', 'ivn', 'mus', 'ld', 'sub', '50', 'sub', 'mg', 'kg', 'mild', 'gastrointestinal', 'side', 'ef']

```
project_demo > dq > out > gold > samples > 📄 train_sample.csv
1  drug_id,drug_name,avg_mass,description,indication,mechanism,toxicity,ingest_dt
2  DB04604,5-iodotubercidin,392.1498,,__MISSING__,,2025-11-04
3  DB00197,Troglitazone,441.54,Troglitazone was withdrawn in 2000 due to risk of hepatotoxicity. It w
4  DB03787,"(5-Hydroxy-6-methyl-4-{[(E)-(3-oxo-1,2-oxazolidin-4-ylidene)amino]methyl}-3-pyridinyl)met
5  DB08186,(3E)-4-(1-METHYL-1H-INDOL-3-YL)BUT-3-EN-2-ONE,199.2484,,__MISSING__,,2025-11-04
6  DB07556,CGS-27023,393.457,,__MISSING__,,2025-11-04
7
```

```
project_demo > dq > out > gold > samples > 📄 test_sample.csv
1  drug_id,drug_name,avg_mass,description,indication,mechanism,toxicity,ingest_dt
2  DB13918,Diclofop-methyl,341.186,Herbicide for control of wild oat & foxtails in cereal crops.,__MI
3  DB18749,Soquelitinib,514.66,,__MISSING__,,2025-11-04
4  DB08414,PNU-142721,306.771,,__MISSING__,,2025-11-04
5  DB06865,6-CARBAMIMIDOYL-2-[2-HYDROXY-6-(4-HYDROXY-PHENYL)-INDAN-1-YL]-HEXANOIC ACID,382.4528,,__MI
6  DB13664,Formocortal,569.06,,__MISSING__,,2025-11-04
```

ment_text.py • 🔄 prepare_splits.py 🔄 silver_to_gold_drugs.py 📄 val_sample.csv ✕ 🔄 config_io.py

```
project_demo > dq > out > gold > samples > 📄 val_sample.csv
1  drug_id,drug_name,avg_mass,description,indication,mechanism,toxicity,ingest_dt
2  DB07041,"N-[2-(2,4-diaminopyrido[2,3-d]pyrimidin-7-yl)-2-methylpropyl]-4-phenoxybenzamide",428.486
3  DB06249,Arzoxifene,475.6,,__RARE__, "Arzoxifene is a selective estrogen receptor modulator (SERM)
4  which antagonizes estrogen in mammary and uterine tissue, but acts as an estrogen receptor agonist
5  ",,2025-11-04
6  DB13618,Ipriflavone,280.3178,,__MISSING__,,2025-11-04
7  DB01981,"3,6-Anhydro-2-(hydrogen sulfate)-alpha-D-galactopyranose",242.204,,__MISSING__,,2025-11-
8  DB08569,"3-PYRIDIN-4-YL-2,4-DIHYDRO-INDENO[1,2-.C.] PYRAZOLE",437.332,,__MISSING__,,2025-11-04
```

```
(datapipeline) spartan@MLK-SCS-FVFGXRHNQ6L4 project_demo % python scripts/llm_enrich_drugbank.py
[INFO] Using latest DrugBank sample: s3://bronze/silver/drugbank/drugbank_sample_20251208_024625_31e554d3df3d4d448f7c68e20e64e675.jsonl
[INFO] Will write enriched output to s3://bronze/gold/drugbank_llm/drugbank_llm_20251208_083920.jsonl
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] Processed 5 records...
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] Processed 10 records...
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] Processed 15 records...
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] Processed 20 records...
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[INFO] Processed 25 records...
[INFO] Wrote 25 enriched records to s3://bronze/gold/drugbank_llm/drugbank_llm_20251208_083920.jsonl
(datapipeline) spartan@MLK-SCS-FVFGXRHNQ6L4 project_demo % aws --endpoint-url http://localhost:9000 s3 ls s3://bronze/gold/drugbank_llm/

2025-12-08 00:36:51      46651 drugbank_llm_20251208_083609.jsonl
2025-12-08 00:39:58      33593 drugbank_llm_20251208_083920.jsonl
```


Data Warehouse Demo

Our robust data warehouse leverages Delta Lake to ensure data integrity, versioning, and high-performance analytics, forming the bedrock for advanced insights.

Delta Lake Storage Layers

Silver and Feature layers are stored as Delta tables (silver/silver_good_delta, feature/drugs_*_delta), providing ACID properties and efficient data management.

Gold Layer Readiness

The enriched Gold JSONL is seamlessly integrated into the warehouse, ready for consumption by downstream analytics, reporting, and AI applications.

Efficient Query Examples

Demonstrates querying capabilities using Spark and Delta Lake, highlighting fast data retrieval and complex analytical operations directly on the structured data.

Seamless Schema Evolution

Delta Lake's built-in support for schema evolution allows for flexible and continuous updates to data schemas without disrupting existing pipelines or requiring costly data migrations.

OBJECT STORE
Community Edition

Create Bucket

Filter Buckets

Buckets

- bronze
- raw

Object Browser

Start typing to filter objects in the bucket

bronze

Created on: Wed, Oct 01 2025 19:32:00 (PDT) | Access: PRIVATE | 1.8 GiB | 8941 Objects

Refresh | Upload

bronze / silver / drugs

Create new path

Name	Last Modified	Size
_delta_log	-	-
part-00000-0bb6645f-7bc1-4e80-a890-614eca002a4c-c000.snappy.parquet	Wed, Nov 05 2025 10:21 (PST)	762.4 KiB
part-00000-1573a9e4-91c5-41d9-b763-54546143177b-c000.snappy.parquet	Thu, Oct 02 2025 02:51 (PDT)	773.9 KiB
part-00000-9a79b85d-acc3-4f9c-bcb4-999dabb445d6-c000.snappy.parquet	Thu, Oct 02 2025 02:58 (PDT)	762.4 KiB
part-00000-c3f3a4fd-c5b1-48df-8f11-85c66b2cdb29-c000.snappy.parquet	Thu, Oct 02 2025 02:44 (PDT)	938.8 KiB

part-00000-0bb6645f-7bc1-4e80-a890-614eca002a4c... (762.4 KB)

#	drug_id	drug_name	avg_mass	description	indication	mechanism	toxicity	ingest_dt
1	DB00005	Etanercept		Dimeric fusion protein consisting of the extracellular ligand-binding portion of the human 75 kiloda...	Etanercept is indicated for the treatment of moderately to severely active rheumatoid arthritis in a...	There are two distinct receptors for TNF (TNFRs), a 55 kilodalton protein (p55) and a 75 kilodalton ...		2025-11-04
2	DB00011	Interferon alfa-n1		Interferon alfa-n1 consists of purified, natural (n is for natural) alpha interferon subtypes, at le...	For the treatment of venereal or genital warts caused by the Human Papiloma Virus.	Interferon alpha binds to type I interferon receptors (IFNAR1 and IFNAR2c) which, upon dimerization,...		2025-11-04
3	DB00012	Darbepoetin alfa		Human erythropoietin with 2 aa substitutions to enhance glycosylation (5 N-linked chains), 165 resid...	For the treatment of anemia (from renal transplants or certain HIV treatment)	Darbepoetin alfa stimulates erythropoiesis by the same mechanism as endogenous erythropoietin. Eryth...		2025-11-04
4	DB00013	Urokinase		Urokinase is an endogenous peptide that is cleaved in the presence of plasmin between lysine 158 and...	In Canada, urokinase is indicated for lysis of acute massive pulmonary emboli, acute thrombi obstruc...	Urokinase is a serine protease. [A191928] It cleaves plasminogen to form the active fibrinolytic prot...	Patients experiencing an overdose may present with bleeding.[L12141] Treat patients with symptomatic...	2025-11-04
5	DB00042	Botulinum toxin type B		Neurotoxin produced by fermentation of clostridium botulinum type B. The protein exists in noncovale...	For the treatment of patients with cervical dystonia to reduce the severity of abnormal head positio...	Botulinum Toxin Type B binds to and cleaves the synaptic Vesicle Associated Membrane Protein (VAMP, ...	One unit of Botulinum Toxin Type B corresponds to the calculated median lethal intraperitoneal dose ...	2025-11-04

Pipeline Monitoring & Management

Airflow acts as the central nervous system for our data pipeline, providing comprehensive orchestration and monitoring capabilities that ensure reliability and transparency.

Real-time Status Updates

Monitor the live status of each task, allowing for immediate identification of bottlenecks or failures.

Detailed Logging & Metrics

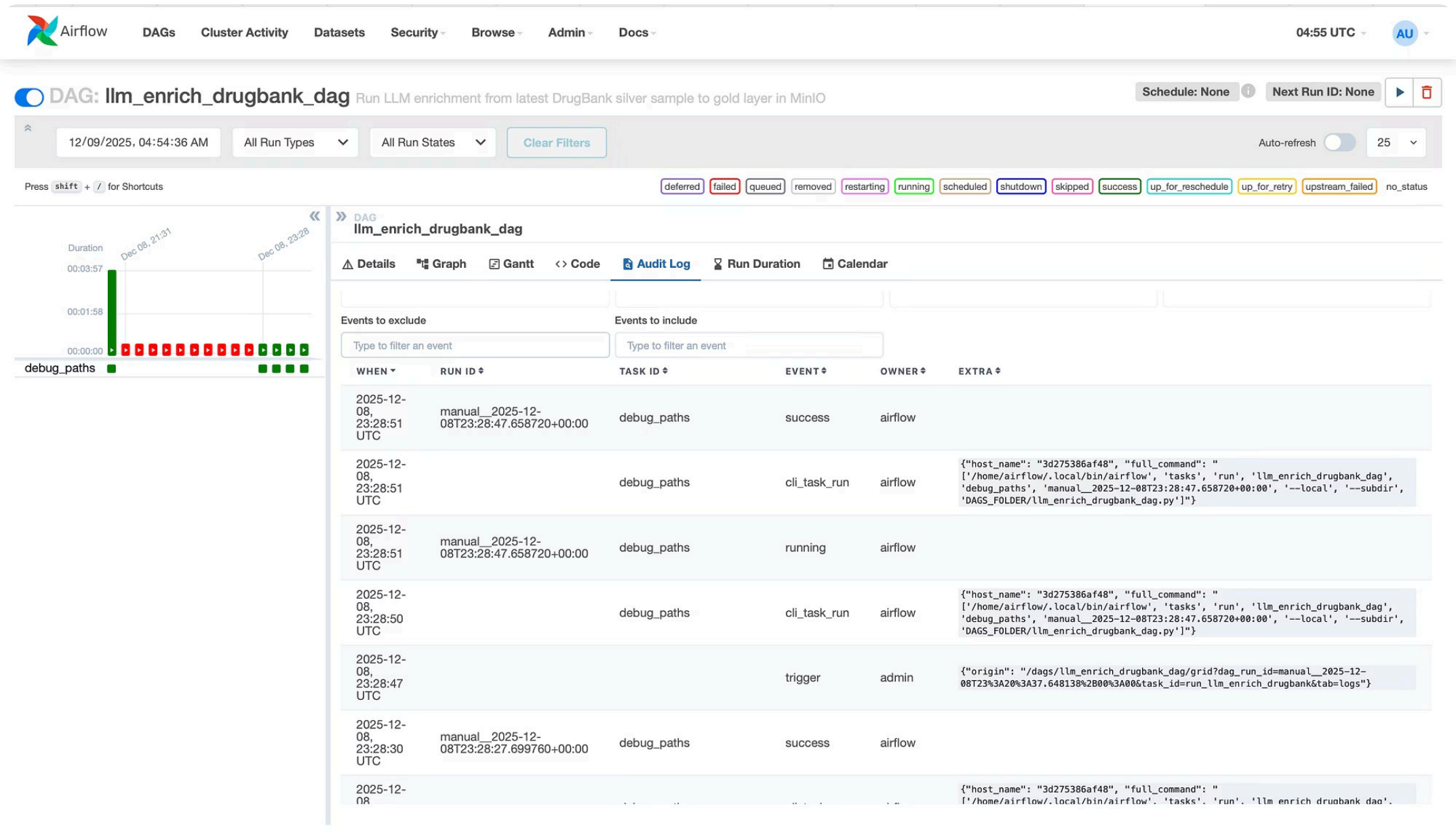
Access granular logs for every step, providing insights into task execution, resource utilization, and potential issues.

Automated Retries & Error Handling

Benefit from Airflow's built-in retry mechanisms and robust error reporting to minimize downtime and ensure data integrity.

Flexible Run Management

Execute DAGs on a schedule or trigger them manually, with full traceability of all past and current runs.



Thank You!

Any questions?